



# Healthy Skepticism Session 1

The five faults, and the first worked example



Murray Cantor, PhD & Joel Keenan, MD [venue and date]



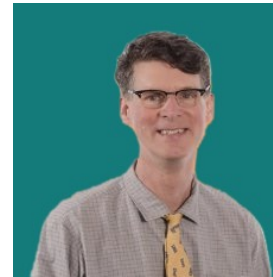
# About Us



**Dr. Murray Cantor**

*PhD, Mathematics — UC Berkeley*

- Not anti-vax or MAHA
- From that perspective, there is a lot of confusion about medical claims
- Respect for the challenge of medical research



**Dr. Joel Keenan**

*MD, Univ of Vermont*

- Practicing physician with direct patient-care experience
- Translates clinical evidence into plain-language guidance
- Shares a commitment to honest, balanced medical communication



# Disclaimer

## **This presentation is for educational purposes only.**

The content does not constitute medical advice and is not a substitute for professional diagnosis, treatment, or care. Nothing presented here should be used to make decisions about your health or the health of others.

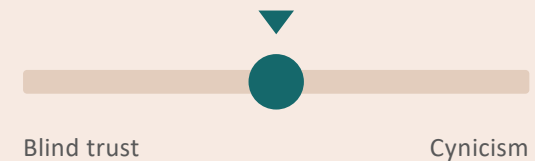
Always seek the advice of a qualified physician or other licensed healthcare provider with any questions you may have regarding a medical condition, test result, or treatment. Never disregard professional medical advice or delay seeking it because of something discussed in this presentation.



# Why the Course

- Empower patients to:
  - Empower patients to have informed discussions with their doctors
  - Make better health decisions
  - Be aware, but not cynical

## Healthy skepticism

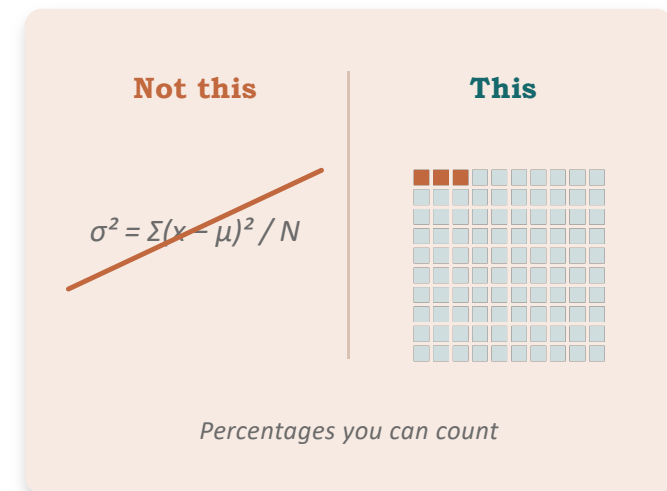


*Ask hard questions — without dismissing good science*



# What to Expect

- We do the math so you don't have to — figures, not formulas
- A lot of real-world examples
- Hopefully, lots of discussion
- All assertions will be backed up
- A companion website with the slides
  - A take-home summary paper for each day
  - Deeper whitepapers on single topics, for the curious





# The Five Faults

1

## Relative reporting

Day 1

*What does a 36% improvement from taking a drug really mean?*

2

## Probability reversal

Day 2

*What does a positive COVID test really mean?*

3

## Arbitrary categories

Day 3

*How concerned should you be if your systolic BP goes from 119 to 120?*

4

## The Bernoulli fallacy

Day 3

*Why does the published medical advice keep changing?*

5

## The causal leap

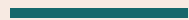
Day 4

*Does bird watching prevent dementia?**Once you see them, you can't unsee them.*



**FAULT 1**

# Relative Reporting

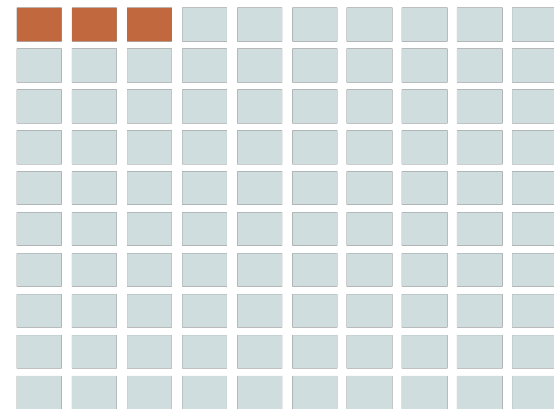


**Relative vs. absolute risk**



# Addressing Uncertainty

- Medicine is uncertain. Every claim is a claim about chances.
- The mathematics of uncertainty is probability theory, which can be abstract and obtuse.
- Fortunately, every concept we need can be explained by counting.
- No equations or formulas. Pictures and a few videos.



*Each square is one person. A hundred squares, a hundred people.*

*Shaded squares are the ones it happens to. Here, 3 in 100.*

*Illustrative. Every idea in this course is a picture like this.*





# “Lipitor reduces Your Risk of Heart Attack by 36%!”

*Why the number in the ad isn't the number that matters to you*

*“In patients with multiple risk factors for heart disease, Lipitor reduces the risk of heart attack by 36%.”*

## 36% of what?

A relative cut, not 36 fewer per 100. It shrinks a small base, from 3.0% to 1.9% over about three years.

## Your baseline is hidden

The headline never states the starting risk, nor that it applies to high-risk hypertensive patients.

## The fine print tells on it

The ad's own asterisk: 3% of placebo patients had a heart attack, versus 2% on Lipitor.

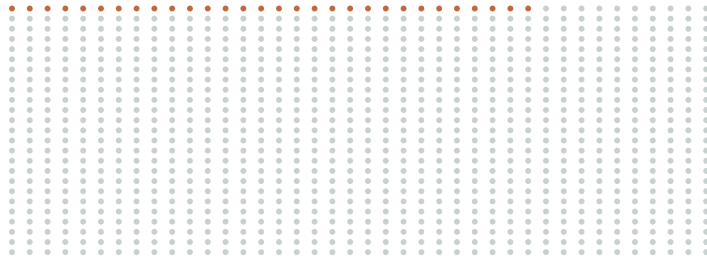
*Sources: actual Lipitor advertisement (Pfizer, c. 2003); ASCOT-LLA (Sever et al., Lancet 2003) and Lipitor prescribing label, 36% relative risk reduction, 1.9% vs 3.0%,  $p = 0.0005$ .*



## The Same 36%: Risk Falls From 3.0% to Just 1.9%

*That 36% cut = 11 fewer heart attacks per 1,000 over ~3 years; 989 in 1,000 see no difference*

Without Lipitor

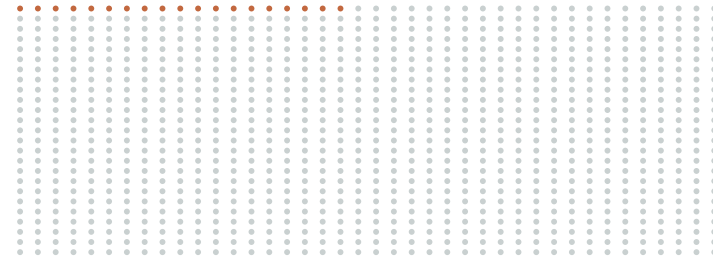


30 in 1,000 have a heart attack

**36% Relative**

the headline number — sounds dramatic

With Lipitor



19 in 1,000 have a heart attack

**3.0% → 1.9% Absolute**

a 1.1-point drop · treat 91 to help 1 (NNT 91)

*Source: ASCOT-LLA, atorvastatin 10 mg vs placebo, 1.9% vs 3.0% over 3.3 years; absolute reduction 1.1%, NNT 91. Lancet 2003; Lipitor label.*

# How to Read the Next “36%” You See

*Three questions that turn any relative-risk headline into a real one*

***“New study: cuts your risk by a third!”***

## **Percent of what?**

A percent with no starting number can't be judged. Find the two actual rates being compared.

## **Find the absolute gap**

Subtract them. That difference, per 100 or per 1,000, is the benefit you would actually get.

## **Then weigh it for you**

Against your own baseline risk, the side effects, and the cost. The headline never does this.

*The method applies to any relative-risk claim. Worked example throughout: Lipitor, ASCOT-LLA (Lancet 2003).*



# What we've learned

One way a true number can mislead you, and a look ahead.



## 1 · Relative reporting

The trap: A big-sounding percentage is usually relative, a tiny absolute change made to sound big.

*Ask: how many real people does it move? (absolute change, NNT)*



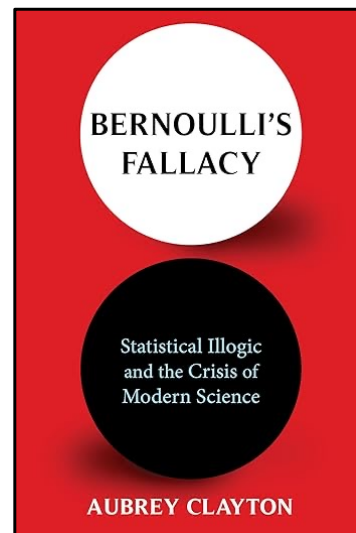
## 2 · Next week: Probability reversal

The trap: A number about a crowd gets read as a probability about you.

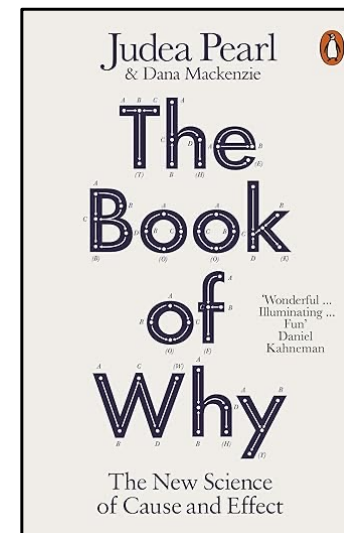
*Ask: given what? What does a positive test actually mean for me?*



## Good Background Texts



*How a 300-year-old logical error fuels today's replication crisis*



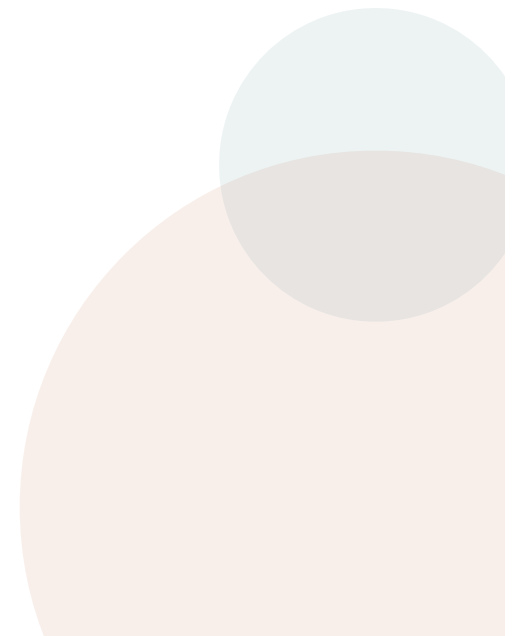
*How to reason about cause and effect*



# Questions, and a conversation

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# Backup Slides

## FAULT 3

# Arbitrary Categories

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**Read the curve, not the cutoff**





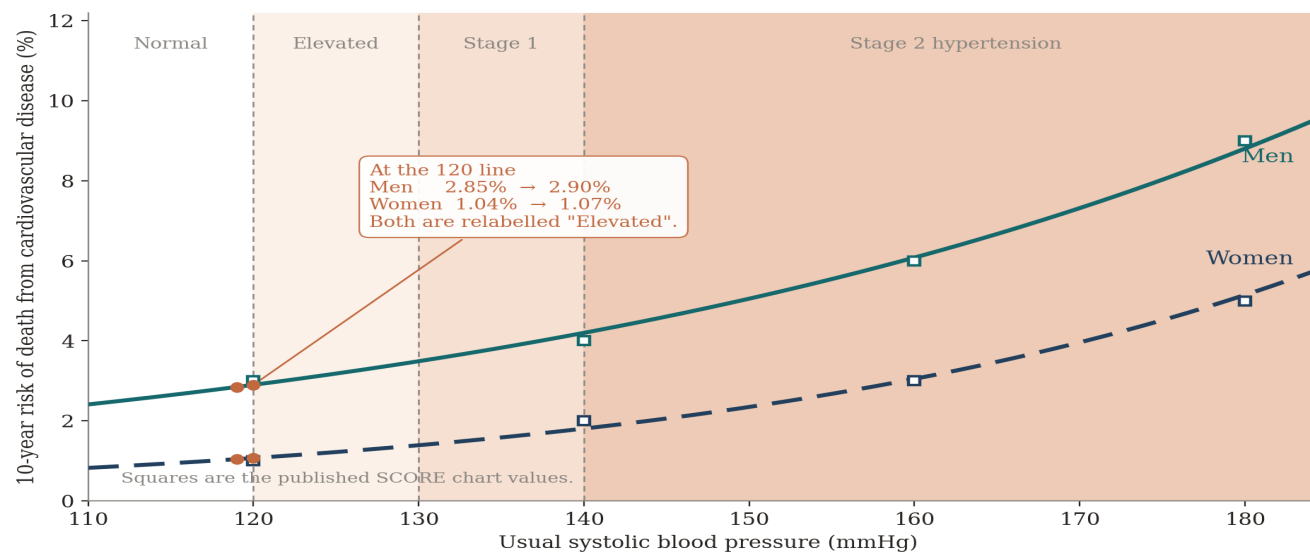
## Where Does “High” Begin?

- Your systolic reads 120. Last year it read 119.
- Nothing about you changed. The label did.
- Every number a doctor measures is continuous. Every label is a box.
- Someone had to decide where the boxes begin, and the curve was not consulted.



# Where the categories fall

The risk curve does not know where the categories are

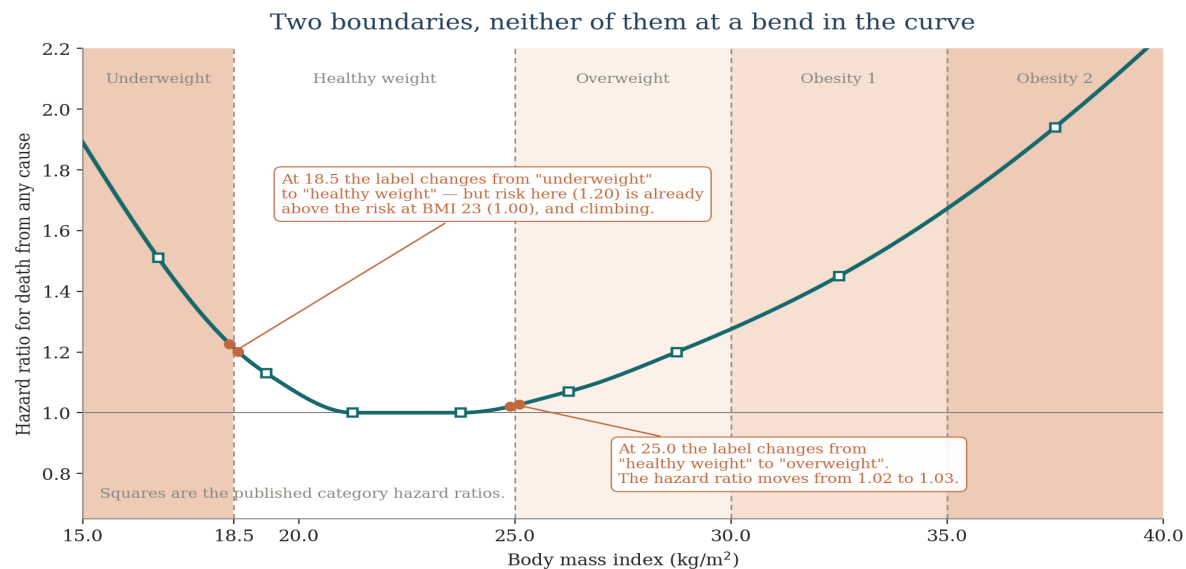


One cutoff, two very different absolute risks. The boundary is the same for everyone; what it means is not.

Squares: SCORE low-risk-region chart, age 65, non-smoker, total cholesterol 5 mmol/L (Conroy et al., Eur Heart J 2003; chart ©ESC 2018). Chart cells are whole percentages, so the low end carries up to half a point of rounding, and the steeper female slope is partly an artefact of it. Curves are log-linear fits to each row, consistent with the smooth no-threshold relationship reported by Lewington et al. (Lancet 2002). Category boundaries: ACC/AHA 2017.



# The Same Problem in BMI



The curve is flat across the middle and steep at both ends. The category lines fall in neither place.

Squares: Global BMI Mortality Collaboration, Lancet 2016;388:776-786. Hazard ratios relative to BMI 22.5 to under 25, plotted at band midpoints, from 3.95 million never-smokers without chronic disease who survived five years, 385,879 deaths across 189 studies. Curve is a shape-preserving interpolation between those points. Above BMI 25 the slope is steeper in men (1.51 per 5 kg/m<sup>2</sup>) than women (1.30). Category boundaries: WHO. The grade 3 band (BMI 40 to 60, HR 2.76) is off the right of this chart.



## The Lines Move

- June 1998: the NIH lowered the “overweight” line from 27.8 to 25. About 25 million Americans became overweight overnight, without gaining a pound.
- 2017: the ACC/AHA lowered “hypertension” from 140/90 to 130/80, reclassifying 31 million Americans, 14% of all adults.
- Neither curve moved. Only the lines did.
- Sources: NIH/NHLBI Clinical Guidelines, June 1998 (contemporaneous press reports put the figure near 25 million); Whelton et al., 2017 ACC/AHA guideline, and Muntner et al.



## So Why Have Categories At All?

- A box is a decision rule. It tells a busy clinic who to look at twice, and it makes a chart readable in three seconds.
- That is a real use, and a good one. The trouble starts when the box is read as a fact about your body.
- A box cannot tell you how much risk you carry, or how much of it you picked up by crossing the line.
- Ask where you sit on the curve, not which side of the line you are on.

# The Same Result, Told Honestly

*What a 36% reduction actually buys the person who takes it*

***“Over about three years, Lipitor prevented a heart attack in roughly 11 of every 1,000 similar patients treated.”***

## The real numbers

30 in 1,000 had a heart attack without it, 19 with it. The drug spared 11 of them.

## Number needed to treat

Treat about 91 such patients for three years to prevent one heart attack (NNT 91).

## Same fact, two faces

36% and 11 in 1,000 are the identical result. One sounds dramatic, one is honest.

Source: ASCOT-LLA, atorvastatin 10 mg vs placebo, 1.9% vs 3.0% over 3.3 years; absolute reduction 1.1%, NNT 91. Lancet 2003; Lipitor prescribing label.



# How to Read the Headline

1

## Baseline risk

About 3 in 100 untreated patients had a heart attack — a 3.0% absolute risk.

2

## What “36%” means

36% OF 3.0% (=1.1%), not 36 points off it — leaving 1.9% risk.

3

## Absolute reduction

$3.0\% - 1.9\% = 1.1$  percentage points — the real individual benefit.

4

## Decision support

Weigh 1.1 points against side effects and cost; higher-risk people gain more.

A “36% risk reduction” only lowers absolute risk from 3.0% to 1.9% — a real gain of 1.1 points, so about 11 of every 1,000 treated actually benefit.

# A Real Headline: A Proposed New Flu Shot

Healthy Skepticism 

*The same lesson as the 36% Lipitor ad — now on a real 2026 trial*

***“New flu shot is 27% more effective for seniors, trial finds”***

*Illustrative headline, representative of the coverage — not a quotation. The 27% is real.*

## **27% of what?**

A relative figure comparing two shots. The base it shrinks is small, so the percentage alone tells you little.

## **More effective than what?**

The yardstick was a standard-dose shot, not the high-dose one adults 65+ are already advised to get.

## **Your decision needs more**

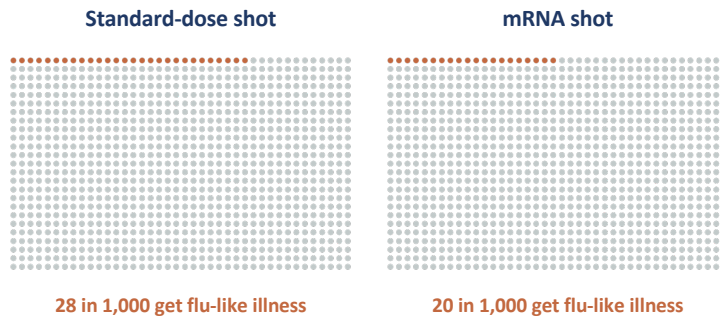
How many people are actually helped, the side effects, and whether any shot beats no shot at all.

Source: Fluent trial (mRNA-1010), New England Journal of Medicine, May 7, 2026 (NEJMoa2516491).



# From 27% to People

What “27% more effective” looks like per 1,000 seniors vaccinated



**8** in 1,000

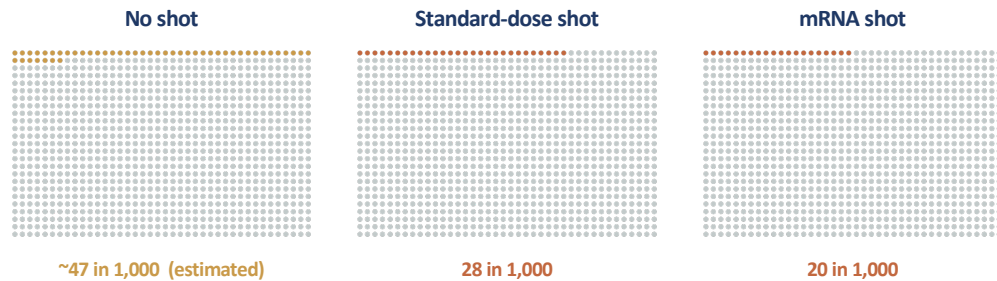
fewer flu-like illnesses with the mRNA shot — the real, absolute gain.

About 125 seniors switched from standard-dose to mRNA to prevent one extra case (NNV  $\approx$  125).

Source: Fluent trial, NEJM 2026 (NEJMoA2516491). Per-protocol flu-like illness 2.0% (mRNA) vs 2.8% (standard-dose).

# The Bigger Question: A Shot, or None?

*The headline compares two shots. The bigger choice is a shot versus nothing.*



≈19 / 1,000

any shot vs none — the big lever

≈8 / 1,000

mRNA vs standard — the headline

**Estimate**

no-shot bar has no trial control; from CDC  
~40% effectiveness in seniors, and it  
swings by season

Sources: mRNA vs standard measured in Fluent trial (NEJM 2026). No-shot column modeled from CDC vaccine-effectiveness for ages 65+ (~40% in 2024-25; 30–41% interim 2025-26).

# Count the Cost, Too

*If the benefit gets an absolute number, so should the side effects*

| Reaction (first week) | mRNA  | Standard | Extra     |
|-----------------------|-------|----------|-----------|
| Sore arm              | 65.8% | 29.8%    | +36 / 100 |
| Fatigue               | 45.1% | 20.3%    | +25 / 100 |
| Headache              | 37.8% | 18.0%    | +20 / 100 |
| Muscle aches          | 35.4% | 11.6%    | +24 / 100 |

## The honest ledger

Most reactions were mild and lasted a couple of days. Serious events did not differ (2.2% vs 1.9%).

The shot prevents roughly 8 to 27 flu cases per 1,000, while adding several hundred short-lived reactions per 1,000.

**Weigh by severity, not by raw count.**

*A sore arm is common but brief; a prevented flu case is rarer but heavier. That trade is the decision.*

Source: *Fluent trial, NEJM 2026* — solicited reactions, first week; serious adverse events 2.2% (mRNA) vs 1.9% (standard).



## Looking Forward, We will Discuss:

1

The state of medical literature

2

Why public health recommendations often flip

3

The importance of low probabilities

4

Causality versus correlation

5

Reading positive test results (COVID, pregnancy)

6

Responding to measurements outside the normal range

7

'Syndrome', 'disorder' and 'spectrum': what doctors mean

# More Effective Than What?

*The comparison behind the headline isn't the shot you'd actually be offered*

## What the headline implies

A new shot that beats your current flu protection by about a quarter.

## What the trial actually tested

The mRNA shot versus a standard-dose shot. Adults 65+ are already advised to get a high-dose or adjuvanted shot, which was not the comparison.

For ages 65 and older, **approval leaned on antibody levels measured against the high-dose shot**, not on a head-to-head count of who actually caught the flu.

***A win over the weaker option is not the same as a win over the option you'd actually get.***

*Source: Fluent trial, NEJM 2026; FDA Vaccines and Related Biological Products Advisory Committee briefing, June 18, 2026.*